MSGR-RAG: Reviewer-Response Report (updated)

Reviewer 2 and Reviewer 3 comments on the revised manuscript. Resubmission deadline: 14 November. This version adds the Component 5 re-run and the co-author's code fixes, and replaces the Table 16 comparison. All numbers come from the result files in *msgr-rag-v1/Datasets/intermediate/*; scripts and outputs are in *msgr-rag-v1/reviewer_response/*. The headline matrix (TP 574, FP 85, FN 0, TN 315; accuracy 91.27%, F1 93.11%) was asserted before any new number was produced and is reproduced by the Component 5 re-run.

1. Summary

- **Core result confirmed by a fresh run.** Component 5 was re-run after the threshold fix: TP 574 / FP 85 / FN 0 / TN 315, accuracy 91.27%, F1 93.11%. The five scores (S_E, S_C, S_D, C(f), Conf(f)) are identical to the old file; only the decision columns changed (Section 2).
- **Threshold selection is not an artefact.** Re-choosing thresholds on a random half and scoring the other half, 20 times, gives 91.2% accuracy (sd 1.1) and 93.0% F1; (0.50, 0.55) was chosen in 20 of 20 splits.
- **The reviewers are right about the test set.** An entity-level lookup (S_E < 1) already scores 100% on all 974 facts, because the domain-inconsistent labels follow the same K=2 rule the domain scale applies. The paper states this and limits its claims to this evaluation setting.
- **The full rule is a conservative operating point.** It misses no hallucination on any of the 974 facts (0 false negatives); a confidence-only rule at the pre-set threshold 0.55 misses 3. The price is 85 true facts routed to review (precision 87.1%), each traceable to a named score.
- **Independent evidence for the multi-scale idea.** In the edge-masking test (no K-derived label), concept and domain scores together reach AUC 0.86 to 0.88 where the entity score reaches 0.33.
- **Against plain RAG, MSGR-RAG is stronger in specific areas, not overall.** Predicate-swapped questions: 39.0% vs 25.3% (p = 0.0025). Domain-inconsistent: 100% vs 94.0% (not significant, n = 50). Overall the two are statistically tied (75.7% vs 78.9%, p = 0.064). On clean triples the core beats a closed-book LLM judge (91.3% vs 61.7%).
- **Two items still decide the outcome:** Table 16 / Fig. 7 must use the primary-run plain-RAG numbers (or document the sensitivity variant), and the gate attribution in Section 5.6 must be rewritten.

2. Component 5 re-run and code fixes (new)

Requested by the co-author (see CODE_FIXES.md). Both fixes were applied to the notebooks; neither changes any score.

Item	Before	After
detect_hallucination defaults	theta_consistency 0.80, theta_confidence 0.50	0.50 / 0.55 (the paper's operating point; already the default in Component 6)
verify_domain_level	Unreachable branch returning 0.6 (Components 5 and 6)	Branch removed in both notebooks. S_D takes only 0.3, 0.5, 1.0 on the 974 facts
Saved decisions	606 HALLUCINATION / 368 VERIFIED; TP 488, FP 118, FN 86, TN 282	659 HALLUCINATION / 315 VERIFIED; **TP 574, FP 85, FN 0, TN 315**
Columns that changed	-	decision (119 rows), level_triggered (248), reason (692). S_E, S_C, S_D, C(f), Conf(f) unchanged
Patterns present	-	I3 on 319 facts, none on 655. I1 and I2 never fire (they need S_D < 0.3, but the lowest mapped S_D is the 0.3 floor)
Files rewritten	-	verification_results.csv, verification_report.json, verification_visualizations.png, and also verification_details.txt

Component 6 was not re-run, as agreed: it already used 0.50 / 0.55 and the removed branch never fired. Nothing has been committed or pushed. This closes the earlier open items 'stale decision column' and 'Table 3 value 0.6 never occurs'.

3. Response status by reviewer comment

Comment	What we did / found	Change in the paper	Status
R2-W1, Q1, S1 / R3 major 1: labels depend on K=2	Confirmed: entity-only = 100%; all 75 domain-inconsistent facts have S_E = 0.1 by construction. Edge-masking test (hide 10-30% of real entity edges, rebuild concept and domain graphs): S_C+S_D AUC 0.86-0.88 vs S_E 0.33.	Sec 4.3, 5.4, 8, abstract: say the 100% checks the set's construction, not unseen-case detection. Add the edge-masking table and paragraph to Sec 5.	Experiment done; text pending

Comment	What we did / found	Change in the paper	Status
R2-W2, Q3, S3: thresholds chosen on the same data	Thresholds came from the sweep on the 981-fact pre-refinement set. Nested 50/50 split x 20: acc 91.2 +/- 1.1, prec 87.0 +/- 1.4, recall 100, F1 93.0 +/- 0.8; (0.50, 0.55) chosen 20/20.	Sec 5.1/5.4: state where thresholds were chosen; add the nested-split row to Tables 9-10. Keep 91.3% as the full-set figure.	Done; text pending
R2-W3, Q2, S2 / R3 major 2: no ablation, confidence-only	Identical-facts ablation (Section 5.1 below): entity-only 100.0%; confidence-only 99.69% (0 FP, 3 FN); full rule 91.27% (0 FN); consistency gate removed: identical; pattern gate removed: 78 FP.	Add ablation table. Rewrite 5.7 around the operating-point trade-off: zero missed hallucinations at a measured cost of 85 flagged true facts; confidence-only is a higher-precision alternative when a miss is less costly. Fix 5.6: pattern I3 alone flags all 85 FPs (78 also fall under the consistency threshold); delete 'non-overlapping subsets'.	Done; text pending
R2-W4, S5 / R3: title, end-to-end claims, readiness	Pipeline: 91.3% on clean triples, 75.7% on the 75.3% resolved, 57.0% over all 974 (UNKNOWN as error).	Drop 'Hallucination-Resilient' from the title; report all three numbers in the abstract; remove 'robust' / 'ready' in 6.5 and 8.	Text pending
R2-W5, Q4, S4: counts do not reconcile	202 candidates = 90 admitted + 112 withheld. Domain graph 119 = 90 K=2 + 29 hand-curated. 149 entities = 95 drugs + 42 conditions + 12 adverse effects (5 cohort drugs yield no relationship). 103,552 relationships collapse to 103,551 edges. 187 to 118 FPs = 36.9% reduction.	Sec 3.3 / Table 4: '119 = 90 + 29 expert-added'. Sec 4.2: explain 95 drugs and the collapsed edge. Sec 5.2: 'nearly halves' becomes '36.9% reduction'. Fix Eq. 14/21 reference; define the satisfiability symbol, divergence, P().	Verified; text pending
R2-W6, S5: refs [14], [20], [35]	Text search of both PDFs. [14]: none of 'multi-scale', 'level of abstraction', 'single level', 'granularity' occurs. [20]: 'granularity' means the retrieved unit; its future-prospects list has no multi-scale verification. [35] is BM25.	Sec 2.2, 1.1, abstract: remove the attributions; rewrite as the authors' own observation ('to our knowledge, the surveyed KG-grounded methods do not perform cross-granularity consistency checks'). Call [35] BM25.	Verified; text pending
R2-W7, S5: weak baseline	Closed-book gemma3 judge on the same 974 triples: acc 61.7%, F1 69.4% vs MSGR 91.3% / 93.1% (McNemar p = 1e-49). TF-IDF (914 aligned facts): 68.6%.	Sec 4.4/5.5: add the judge row; state it is closed-book and one small model; list NLI, RAGAS, ARES, RAGTruth as future work.	One baseline done; others future
R3: NL degradation, 5-sentence regex test, inconsistent denominators	Plain RAG and MSGR recomputed on the same 733 resolved questions and on all 974 (Section 4).	Replace Table 16 with the aligned table (LaTeX in Appendix A). Label Sec 6.7 'anecdotal, n = 5'. Attribute the predicate-swap drop to question generation and list it as a pipeline limitation.	Done; text pending
R3 minor: abstract length; deterministic vs learned verifier; generalisation	No experiment needed.	Cut the abstract to about 150 words. Scope paragraph in Sec 1.3. Label Eq. (5) 'not evaluated'. Sec 8: one DrugBank cohort, one LLM.	Text pending

4. MSGR-RAG versus plain RAG: where it excels

Plain RAG is the primary measured run (*simple_rag_v2_results.csv*): same gemma3 model, same 974 questions, the entity's first five known facts as context, and the model's own VERIFIED/HALLUCINATION label. Both systems are scored on the 733 questions the MSGR pipeline resolves. Significance is an exact McNemar (sign) test on discordant pairs.

Category (resolved subset)	n	MSGR-RAG	Plain RAG	MSGR right / plain wrong	Plain right / MSGR wrong	p-value	Reading
Predicate-swapped	154	39.0%	25.3%	33	12	0.0025	MSGR better, significant
Domain-inconsistent	50	100.0%	94.0%	3	0	0.25	MSGR higher; not significant (small n)
Absent relationship	170	98.2%	100.0%	0	3	0.25	Tie
Real relationship	359	77.4%	89.7%	23	67	3.8e-6	Plain RAG better, significant
Overall	733	75.7%	78.9%	59	82	0.064	Statistically tied
All 974 (UNKNOWN = error)	974	57.0%	79.8%	-	-	-	Coverage cost of abstaining

What can honestly be emphasised

- **Predicate-swapped facts** (the patient-safety error, for example 'causes' for 'treats'): significantly better than plain RAG in the pipeline (39.0% vs 25.3%, p = 0.0025), and on clean triples the verification core catches 100% against 83.0% for the closed-book judge and 0.0% for TF-IDF. The low absolute pipeline figure comes from the question generator drifting to an interaction phrasing (91 of 94 resolved errors, paper Sec 6.4); that limit applies to both systems.

- **Domain-inconsistent facts:** 100% vs 94.0%, same direction, but 3 versus 0 discordant pairs on 50 questions is not significant. Report as 'higher, not significant'.
- **Zero missed hallucinations** on the clean-triple core, with an auditable reason for every verdict (S_E, S_C, S_D, C(f), Conf(f), pattern).
- **Abstention with a traceable reason:** UNKNOWN on 24.4% of questions (entity outside graph coverage). Plain RAG scores 82.6% on those 241 questions, so coverage, not correctness, is the price.
- **State alongside:** lower overall accuracy than plain RAG, significantly lower on real relationships (77.4% vs 89.7%), 57.0% over all 974 when UNKNOWN counts as an error (plain RAG 79.8%). Plain RAG here retrieves the first five facts of the subject, not question-relevant ranking, so it is weak but fair by construction; say so.

Exact edits to the paper for this comparison

Where	Edit
Abstract	Replace 'outperforms a plain retrieval-and-generation baseline on every category, most substantially on domain-inconsistent facts (+19.5 points)' with: 'Against a plain RAG baseline on the 733 resolved questions, MSGR-RAG is significantly more accurate on predicate-swapped questions (39.0% vs 25.3%, p = 0.0025), higher but not significantly so on domain-inconsistent questions (100% vs 94.0%), and less accurate on real relationships (77.4% vs 89.7%); overall the two are statistically tied (75.7% vs 78.9%, p = 0.064).'
Sec 1.3	Say 'compare against a plain retrieval-and-generation baseline, identifying the categories where verification helps and where it costs accuracy'.
Sec 6.6, Table 16, Fig. 7	Replace with the aligned table (Appendix A) and the all-974 row. Delete 'improves accuracy on every category'. Redraw Fig. 7 from the same numbers; the old figure plots the 67.8% baseline.
Sec 6.5	Delete 'robust'. Say: 'on resolved questions the core's category pattern is preserved for three of four categories; the exception is predicate-swapped (39.0%), caused by question generation.'
Sec 8 (Conclusion)	Replace 'improves over a plain RAG baseline on every category' with the abstract sentence; add one sentence on abstention (24.4% UNKNOWN) and traceability.
Sec 7 (limitations)	Single LLM (gemma3), baseline retrieval is the first five entity facts, one DrugBank cohort, small n in the domain-inconsistent category (50).

5. Evidence added in this revision

5.1 Ablation (identical 974 facts, thresholds 0.50 / 0.55)

Rule	FP	FN	Accuracy	Precision	Recall	F1
Entity-only (S_E < 1)	0	0	100.0%	100.0%	100.0%	100.0%
Confidence-only (0.55)	0	3	99.69%	100.0%	99.48%	99.74%
Confidence-only (gap midpoint 0.687, set after seeing labels)	0	0	100.0%	100.0%	100.0%	100.0%
Full rule (C | pattern | Conf), paper	85	0	91.27%	87.10%	100.0%	93.11%
Full minus consistency gate	85	0	91.27%	87.10%	100.0%	93.11%
Full minus pattern gate	78	0	91.99%	88.04%	100.0%	93.64%
Concept-only (S_C < 1)	7	123	86.65%	98.47%	78.57%	87.40%
Domain-only (S_D < 1)	111	145	73.72%	79.44%	74.74%	77.02%

AUC on the 974 facts: S_E 1.000, S_C 0.887, S_D 0.715, C(f) 0.807, Conf(f) 1.000. Source: *out/ablation.csv*.

5.2 Edge masking (independent of the K=2 rule)

Hide a random fraction of the 400 real entity edges, rebuild the concept and domain graphs from the masked entity graph (same K=2 rule plus the 29 hand-curated edges), and ask whether the scores separate the held-out TRUE facts from the 574 false facts among facts the entity scale cannot support. Five seeds per rate. The scorer is a re-implementation validated against the frozen scores (S_E 100%, S_C 99.9%, S_D 97.5% agreement).

Held out	AUC S_E	AUC S_C	AUC S_D	AUC S_C+S_D	AUC Conf	AUC C(f)	Held-out true flagged by full rule
10%	0.328	0.828	0.674	0.856	0.752	0.116	100%
20%	0.330	0.835	0.704	0.876	0.772	0.101	100%
30%	0.331	0.841	0.698	0.880	0.770	0.098	100%

Reading: concept and domain scores carry real signal where the entity scale is blind. The shipped rule does not use it: C(f) is anti-correlated with truth here (AUC about 0.10) and the full rule flags every held-out true fact. A crude accept rule (S_C = 1 and S_D = 1) recovers 62-63% of held-out true facts at a 5.2% false-accept rate; present this only as exploratory and as the main future-work direction.

5.3 Baselines on the same 974 clean triples

System	n	Accuracy	Precision	Recall	F1
MSGR-RAG core (full rule)	974	91.3%	87.1%	100.0%	93.1%
Closed-book gemma3 judge (temperature 0)	974	61.7%	65.6%	73.5%	69.4%
TF-IDF similarity (pre-refinement set, aligned)	914	68.6%	100.0%	44.2%	61.3%

Per-category accuracy (MSGR / judge / TF-IDF): absent 100 / 68.6 / 73.6; predicate-swapped 100 / 83.0 / 0.0; domain-inconsistent 100 / 68.0 / 46.7; real 78.8 / 44.8 / 100. The judge is closed-book (no graph evidence) and one small model: it shows the task is not solved by a general LLM, not that MSGR beats a retrieval-grounded judge.

6. Items still open before resubmission

Item	Why it matters	Action	Owner
Provenance of the Table 16 plain-RAG column	The paper's 67.8% (84.4 / 71.6 / 80.5 / 30.6) equals column simple_rag_decision in plain_rag_sensitivity.csv, which differs from the primary run in 117 decisions with identical answer text; no code in the repository produces it. A reviewer who reruns plain RAG gets 78.9%.	Report the primary run in Table 16. If the sensitivity variant stays, document its procedure and report it separately.	Author
Gate attribution (Sec 5.6)	Pattern I3 alone flags all 85 FPs; '78 + 7 non-overlapping' depends on check order.	Rewrite as in Section 3.	Author
I1 and I2 never fire	Both need S_D < 0.3 but the lowest mapped S_D is the 0.3 floor. Only I3 appears (319 facts). Making the test '<=' would change results, so the rule stays as published.	Say in the paper that I1 and I2 do not fire on this set.	Author
interacts_with reverse-direction rule	S_E and S_C check the reverse edge only for an exact predicate match; a reverse edge with another predicate gets the floor (0.1 / 0.2). My re-implementation scores it as a mismatch. 24 S_D cases remain unexplained by this note.	Make the Table 3 caption match the code. Confirm the S_D rule.	Co-author
Deduplication (Components 1 and 2)	Both read drugbank_clean.csv; the deduplicated file is used only by the ATC mapping. The paper says cleaning is upstream of all construction.	Co-author has changed the paper sentence; do not change the components.	Done
One concept per entity (Eq. 3)	entity_to_concept is a plain dictionary, while Eq. 3 allows a set of concepts. 53 of 95 drugs map to the catch-all concept Other_Drugs.	Reword Eq. 3 or mention the single-concept mapping and the catch-all.	Author
Domain-inconsistent vs S_D	15 of 75 such facts have S_D = 1.0 under the ATC-grounded mapping (category defined under the transitive mapping).	State the definition precisely in Sec 4.3.	Author
Cohort and cleaning wording	The old rebuild notes list 17 vs 20 duplicated cohort drugs and area labels that do not match drug types; not verified in msgr-rag-v1.	Check Sec 4.1 wording against the data before resubmitting.	Author

7. Future scope (accepted as outside this revision)

- Use the concept/domain signal in the decision rule: a calibrated or learned combination validated on held-out edges is the main way to demonstrate a multi-scale gain.
- Learned verifier (Eq. 5) with negative sampling independent of the evaluation-set construction rules.
- Independent labels for domain-inconsistent facts (expert review or a clinical ontology other than the K=2 rule).
- Stronger baselines: NLI faithfulness models, RAGAS, ARES, RAGTruth-trained detectors, a retrieval-grounded LLM judge, plain RAG with question-relevant retrieval.
- Generalisation: a second LLM, a second domain or dataset (for example SIDER, BioASQ, PubMedQA), multi-hop questions.
- Extraction from genuine text with a parser- or model-based extractor, evaluated on more than five sentences.
- Sensitivity studies for the Table 3 floors, the confidence weights, and K.

8. Suggested order of work to 14 November

Step	Task	Effort
1	Document or replace the plain-RAG sensitivity column; replace Table 16 and Fig. 7 (Appendix A).	half a day
2	Rewrite Sec 5.6, 5.7, 4.3, 5.4 and the abstract and conclusion (Sections 3 and 4).	1 day
3	Mechanical fixes: title, Sec 3.3 / Table 4 counts, Eq. cross-reference, symbol definitions, refs [14] [20] [35], abstract length, I1/I2 note, Eq. 3 wording.	2-3 hours
4	Add three tables to Sec 5 (nested split, ablation, edge masking) and the judge row.	half a day

Step	Task	Effort
5	Write the response letter point by point from Section 3 (concede circularity and the confidence-only result; cite the new tables).	half a day
6	Optional: rerun plain RAG into a new file and compare with the primary run; confirm the S_D reverse-direction rule.	1-2 hours each

Appendix A. Replacement for Table 16 (LaTeX)

```
\begin{table}[t]
\centering
\caption{Plain RAG vs.\ full pipeline + MSGR-RAG on the same 733 questions the
pipeline resolves (accuracy, \%). The last row scores both systems on all 974
questions, counting UNKNOWN as an error. Plain RAG is the primary run
(\texttt{simple\_rag\_v2\_results.csv}). On the resolved subset the overall
difference is not significant (McNemar $\chi^2=3.43$, $p=0.064$).}
\label{tab:plain_rag_vs_msgr}
\begin{tabular}{|l|c|c|c|}
\hline
\textbf{Category} & \textbf{n} & \textbf{Plain RAG} & \textbf{Full pipeline + MSGR-RAG} \\
\hline
Overall (resolved)    & 733 & \textbf{78.9} & 75.7 \\ \hline
Absent relationship   & 170 & \textbf{100.0} & 98.2 \\ \hline
Real relationship     & 359 & \textbf{89.7} & 77.4 \\ \hline
Domain-inconsistent   & 50  & 94.0 & \textbf{100.0} \\ \hline
Predicate-swapped     & 154 & 25.3 & \textbf{39.0} \\ \hline
\hline
Overall (all 974, UNKNOWN = error) & 974 & \textbf{79.8} & 57.0 \\ \hline
\end{tabular}
\end{table}
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Appendix B. Reproduction

From *msgr-rag-v1/reviewer_response/*: *python3 -I analyses.py* (ablation, nested split, aligned denominators, count reconciliation); *python3 -I edge_masking.py*; *python3 -I judge_eval.py* (judge predictions are saved; llm_judge.py regenerates them in about 10 minutes). Outputs are in *out/*. Component 5: run *notebooks-drugbank/MSGR-RAG-Component-5-Fact_Verification_System.ipynb* top to bottom (about 7 seconds); it rewrites the verification_* files. The per-category plain-RAG significance test was computed from component6_e2e_results.csv and simple_rag_v2_results.csv. Narrative notes: FINDINGS.md and MODIFICATION_REPORT.md.